

Sishir Phuyal

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SUMMARY

Software engineer with 4 years at a life insurance carrier, from intern to full-time, where I modernize legacy systems and build the financial engines behind agent compensation. I also ship full-stack AI products end to end, including a live Google Play app and an AI lecture platform.

TECHNICAL SKILLS

Languages: TypeScript, C#, JavaScript, SQL, Python

Backend & Data: .NET, Node.js, Express, Django, PostgreSQL, SQL Server, Prisma, Knex, REST APIs

Frontend & Mobile: React, Next.js, React Native, Expo, Tailwind CSS, Vite

AI & Cloud: OpenAI API, Groq, Anthropic Claude, AWS Bedrock, Vercel, Stripe

Practices: Git, CI/CD, Monorepos (Turborepo), Automated Testing, X12 EDI

EXPERIENCE

Investors Heritage

Louisville, KY

Life insurance carrier supporting 15,000+ agents and 300,000+ policies. I build the internal .NET platform behind agent servicing, policy administration, and finance.

Software Developer

May 2025 – Present

Software Developer Intern

May 2022 – May 2025

- Rearchitected the backend data-access layer into a centralized database class and dynamic query builder, replacing scattered raw SQL with composable, maintainable methods (offset/fetch pagination, dynamic filtering); adopted as the **company-wide standard**, now used by **10+ developers** across multiple teams. I documented it in code and in a team wiki guide, and onboarded other developers onto it.
- **Proposed and delivered** the fix that cut load times on the heaviest search pages from **five to ten minutes to under ten seconds**. Large agency accounts hold thousands of policies, and those grids were fetching every record before rendering a single row; I moved them to server-side pagination so the backend returns one page at a time.
- **Migrated a legacy COBOL program to .NET**, retiring a system the team could no longer maintain. Rebuilt the agent hierarchy workflow that governs how policies are credited, including rules for inserting agents into existing hierarchies and a split-hierarchy path; business users now make changes that previously required a developer.
- Built the **quarterly agent bonus engine**, replacing **days of manual finance work** every quarter. It evaluates chargebacks, lapsed policies, and tier rules across several finance databases to compute bonuses for **thousands of agents**, then generates the annual true-up table at Q4 close.
- Built a one-click simulation on the test environment that replays a full year of chargeback and tier calculations, so finance and agents could verify their own numbers without a developer. I am the team's **go-to for bonus compensation and agent hierarchy**.
- Designed and deployed the scheduled batch jobs behind policy servicing, including a monthly void-policy report emailed to agents that has run in production for **2+ years**.

PROJECTS

Chalk: AI Lecture Capture Platform

Solo · 2026

TypeScript, Next.js 16, React 19, Groq + OpenAI, Neon Postgres, Vercel Blob, Stripe, LTI 1.3, Expo · chalkrecap.com

- Built and launched a serverless media pipeline that turns a recording, upload, or video link into navigable chapters, notes, quizzes, and a tutor citing clickable timestamps, across **82 API routes** serving a web app and an Expo client.
- Engineered the pipeline to survive platform timeouts: audio is chunked and each chunk checkpointed to Postgres, with an atomic claim lock and a cron sweeper reviving runs killed by the 300-second function ceiling, so a **crashed job resumes without re-paying** for transcription.
- Cut transcription cost **roughly 10x** by routing it to Groq behind an OpenAI-compatible fallback and tiering models by task; wired Stripe metered billing, LTI 1.3 launch from Canvas/Moodle, and Zoom/Webex OAuth import.

RoamKit: Offline-First Travel Toolkit

Solo · 2026

React Native, Expo, Expo Router, RevenueCat, EAS, AsyncStorage

- **Shipped to the Google Play Store** after a 14-day closed beta and production review: one Expo codebase deploying to iOS, Android, and web, with 10 utilities that all function with zero connectivity, monetized with RevenueCat in-app purchases.

RayHealth EVV: Medicaid Compliance Platform

2-engineer team · 2026

TypeScript, React, Node.js, Express, PostgreSQL, Expo, Turborepo, AWS Bedrock · rayhealthevv.com

- **Sole designer and developer** of the caregiver mobile app (Expo, iOS and Android) inside a **4-workspace Turborepo** shared with one other engineer: every screen from clock-in through documentation, e-signature, training, mileage, and identity capture.
- Built **GPS visit verification**: the server checks location against the client's registered geofence and **flags any caregiver who moves outside the radius** for review. An offline queue records a clock-in in a dead zone and replays it in order.
- Built the **claims and billing transport**: 837P X12 claim submission with automated 835 remittance ingest, plus scheduled visit submission to state Medicaid aggregators (Sandata, HHAEExchange).

EDUCATION

University of Louisville, J.B. Speed School of Engineering

Aug 2021 – May 2025

Bachelor of Science in Computer Science · Louisville, KY